AI Model Inference and Optimization Techniques

Artificial Intelligence models generally work in two major phases: training and inference. Training is the process in which the model learns patterns from large datasets by adjusting its internal parameters. This phase requires high computational power and is usually performed using GPUs because training involves processing huge amounts of data repeatedly. Inference, on the other hand, is the phase where a trained model is used to make predictions on new input data. For example, when a sentiment analysis model predicts whether a sentence is positive or negative, it is performing inference. Compared to training, inference requires fewer resources and can often be executed on CPUs.

Model inference is an important aspect of real-world AI applications because deployed systems mainly use already trained models to generate outputs. However, running large AI models on systems without GPUs can be challenging due to slower processing speed and higher memory usage. To address these issues, several optimization techniques are used to improve inference performance in non-GPU environments.

One of the most widely used optimization techniques is quantization. Quantization reduces the numerical precision of model parameters, for example converting 32-bit floating-point values into 8-bit integers. This reduces model size, decreases memory usage, and improves inference speed while maintaining acceptable accuracy. Quantization is especially useful for running AI models on CPUs, mobile devices, and edge systems where hardware resources are limited.

Another important technology is ONNX (Open Neural Network Exchange). ONNX is a standardized format used for representing and deploying machine learning models across different platforms and frameworks. It allows models trained in one framework to be optimized and executed efficiently in another environment. ONNX improves portability, reduces dependency issues, and provides faster inference performance, particularly in CPU-based systems.

GGUF is another model format mainly used for Large Language Models (LLMs). It is commonly associated with tools such as llama.cpp and Ollama for running models locally. GGUF is designed to optimize memory usage and improve inference efficiency, making it suitable for executing LLMs on CPU systems without requiring high-end GPUs.

In this project, AI model inference was explored using Hugging Face transformer models in Google Colab. CPU and GPU execution were compared to understand performance differences. It was observed that GPU-based inference was significantly faster, while CPU-based inference was slower but still functional. The study highlights the importance of optimization techniques such as quantization, ONNX, and GGUF in enabling efficient AI model deployment in non-GPU environments.